

FINDABILITY SCIENCES PVT LTD

ARTIFICIAL INTELLIGENCE AND AGRITECH INNOVATION CENTER-PRIMARY SCREENING

Application Id: AIAI-974358

Problem Statements

Climate Resilience

Choose Application Track (Please review [Link])

Track 2 (Pilot and Validation)

Name

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Contact Number of the Authorized Person

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Startup Name

Findability Sciences Pvt Ltd

Type

Startup

Startup(Please Mention)

Growth

Founders Background (2-3 lines)

Anand Mahurkar is the Founder & CEO of Findability Sciences, a Boston-headquartered enterprise AI and data analytics company that helps global organizations turn complex data into actionable insights. Originally from a small town in interior Maharashtra, India, he's a first-generation entrepreneur and immigrant to the U.S., with a background in mechanical engineering and over 30 years of experience leading technology and data-driven business solutions.

Employees

No. of Full Time Employees	120
No. of Contractual Employees	15

Year of Incorporation

2009-08-11T04:00:00.000Z

CIN/Registration No.

U74999MH2009PTC194842

GSTIN (if applicable)

27AABCF3906G1ZV

Startup India / DPIIT Recognition No. (if any)**UDYAM Registration (if any)****State**

Maharashtra

Registered Address

A102, Gala Quest, Paranjape B Scheme, Road No 1, Vile Parle East, Mumbai - 400057

Operational Address in Maharashtra (if different)**Website**<https://www.findability.ai>**PAN of Organization**

AABCF3906G

Prior Govt. Collaboration (Also mention significant non-government collaboration) if any**Financials**

Field	FY (24-25)	FY (23-24)	FY (22-23)
Net Revenue	20.69 (INR crore)	17.27	18.4
Expenses	20.28	18.14	0.02
Net Profit	0.43	-0.87	17.21
Cash Reserves	1.27	1.15	1.73
Avg. Burn Rate	-	-	-
Funding raised till date	0.83	0.53	0.53
Unit economics (e.g. cost per farmer serviced/revenue per transaction)	-	-	-
Committed pipeline or LOIs (if there are al-	-	-	-

ready customers/pi-
lots' agreements)

Upload financial documents (Past 3 years, Audited Balance Sheet, P&L Statement etc)

<https://api-legacy.accubate.app/document/aiaic/7a0bdbab-1056-40e5-9011-0430c030aebd>

Project/Product Name

Stoma Sense: AI-Driven Remote Sensing, Moisture, Growth and Yield Intelligence for Sugarcane in Maharashtra

Solution Synopsis

Stoma Sense is an integrated GEOAI platform that helps farmers track crop growth and optimise sugarcane harvest quality and productivity. It combines Sentinel-1/2 time-series, deep-learning cloud-free reconstruction, cane-moisture samples, yield history, irrigation telemetry and weather data to deliver continuous, plot-level intelligence. Its U-Net GAN engine reconstructs clean multispectral bands even during monsoon cloud cover, ensuring uninterrupted NDVI, NDRE, NDWI, HYDRO+, growth, Potential Growth Index (PGI), weed and stress indicators.

A key differentiator is the cane-moisture model, which learns relationships between spectral indices, rainfall-ET balance, soil/climate/variety/irrigation factors and lab-measured moisture %, enabling accurate moisture estimation and its impact on TCH yield. PGI benchmarks each plot's spectral trajectory against historical cohorts to detect stagnation or hidden stress that NDVI may miss.

Building on this foundation, Stoma Sense forecasts moisture deficit, weed outbreaks, growth anomalies and TCH yield using ML models with SHAP-based explainability. Advisories are delivered via a simple field app, while managers use a web dashboard for planning and monitoring. The modular, cloud-native system scales across mills and districts and adapts to other irrigated crops, providing a blueprint for statewide digital agriculture transformation.

Target Beneficiaries (Class of Beneficiaries)

Farmers

Proposed Project Districts in Maharashtra.

PuneChhatrapati Sambhajinagar

Proposed Project Taluka in Maharashtra - Pune

Baramati

Proposed Project Taluka in Maharashtra - Chhatrapati Sambhajinagar

Chhatrapati Sambhajinagar

Proposed Project Duration (months)

12

Total Project Cost (INR) (AIAIC support + Own Funding/Private Funding)

31800000

Unique Value Proposition of the solution (What value does your solution deliver to your target customer ?)

Stoma Sense unlocks the intelligence embedded in each farmer's land. By integrating multi-season historical data with high-frequency satellite imagery, weather forecasts, irrigation telemetry and field observations, the platform builds a hyper-local, personalized GEOAI model for every plot. Its deep-learning cloud-removal engine reconstructs clean multispectral bands even during monsoon, delivering uninterrupted NDVI+, HYDRO+, growth-curve deviation and stress insights - beyond what conventional NDVI dashboards can offer.

This personalized intelligence supports actionable decisions: when to irrigate, where weeds are emerging, which zones are deviating from optimal growth, and when the crop is approaching harvest readiness. Farmers and factories benefit through improved yield stability, reduced input use, and a scientific understanding of how each field responds to climate, water and management practices.

Product Slides Link: <https://findabilitysciences-my.sharepoint.com/:p:/p/msagar/IQC2PaHZ5AFNQq-qptCuS-BYQAdZWQMwXO6voBcKgEbAQAzg?e=tDk96q>

Does the product have the potential to create employment ?

It can create jobs directly through field officers, agronomy assistants, data annotators, and irrigation technicians, and indirectly by boosting demand for local agri-services, improving efficiency, and strengthening the rural agricultural ecosystem.

Core Technology Stack used/Innovation

AI /ML(Artificial Intelligence/Machine Learning)Remote Sensing Computer VisionGIS NLP (Natural Language Processing) Edge Computing

Explain the technology you are using in detail

Stoma Sense is built on a GEOAI architecture that integrates satellite analytics, agronomic modeling and cloud-native automation into a single operational pipeline. The system runs a multi-stage chain—ingestion of Sentinel-1/2 imagery, atmospheric correction, tiling, cloud-reconstruction inference, spectral index computation and plot-level aggregation—all executed on a scalable cloud environment to support continuous monitoring across large command areas.

Its deep-learning reconstruction module, trained on paired cloudy and cloud-free imagery, restores per-band spectral information, ensuring temporal continuity even when optical data is obstructed. Pixel-level values are transformed into agronomic indicators using rule-based and ML methods. Plot boundaries, variety maps and irrigation metadata contextualize these signals so that predictions reflect real management conditions. Data fusion of radar, optical, moisture samples, telemetry and weather feeds the cane-moisture model, growth segmentation and TCH forecasting.

The platform uses containerised microservices for preprocessing, modeling and alert generation, with autoscaling and asynchronous orchestration so each new satellite acquisition triggers updated analytics and advisories. Stoma Sense's mobile and web layers access results via secure APIs, enabling real-time advisories, field reporting and zone-level dashboards without exposing any personal data.

Technology Readiness Level (TRL) is a measure used to assess the maturity of a particular technology on a scale from 1 to 9

TRL 9 (Technology proven and used in a real industry setting)

Does your project incorporate any open-source technologies, standards, or contributions? If yes, could you specify which ones and how they are used?

Yes. Stoma Sense extensively leverages open-source geospatial, remote-sensing and machine-learning technologies across its full pipeline. Satellite imagery is accessed through Copernicus Dataspace and processed using the openEO API, an open geospatial standard that enables scalable, cloud-agnostic processing of Sentinel-1/2. Cloud-free reconstruction is implemented in PyTorch, using open-source architectures (U-Net, PatchGAN)

adapted to 14-band Sentinel-2 data. Preprocessing and band-level operations rely on GDAL, Rasterio, xarray, rioxarray, NumPy, Pandas and scikit-learn, all of which are open-source scientific libraries. The system adheres to OGC-compliant formats such as GeoTIFF and Cloud-Optimized GeoTIFF, ensuring interoperability with government GIS systems. Vegetation and water indices (NDVI, NDRE, NDWI, EVI, SAVI) follow internationally published open formulas. These open-source components ensure transparency, reproducibility, and long-term sustainability, while Stoma Sense adds proprietary GEOAI layers (moisture model, yield predictors, stress segmentation, potential growth index) on top of this open technological foundation.

Data Sources Used

Geospatial and Satellite Data (Land/crop mapping, drought monitoring)

Describe how the solution ensures compliance with data governance and privacy requirements under DPDP Act, 2023

Secure Data Storage (Encryption at Rest and Transit)

In your business model or technology, which of the following practices do you use to evaluate model safety and ensure reliable and fair outcomes?

Biases (Identifying and mitigating unfair biases in the model)

IP Status

Background IP

Revenue Model

Subscription fees

Target Market/Customer Segment (Intended Customers/Beneficiaries such as Farmers, FPOs, Food Processors, Agri Input Dealers, Agrifintech, etc)

Stoma Sense is designed primarily for farmers, particularly those cultivating sugarcane in water-stressed or climate-variable regions who need timely, plot-level intelligence to manage irrigation, detect stress early, and optimise yield. The platform also serves organizations that work closely with farmer groups—such as FPOs, cooperatives, sugar mills, NGOs, and agri-extension programs—by providing continuous monitoring, operational insights, and decision-support tools for advisory teams. These stakeholders rely on accurate field intelligence to plan interventions, improve grower engagement, and enhance overall productivity. Together, individual farmers and their supporting institutions make up the core beneficiaries and customers of Stoma Sense.

Market size (Total Addressable Market, Serviceable Available Market, Serviceable Obtainable Market), if any

For Sugarcane Crop: In Maharashtra, approximately 3.67 million acres are under sugarcane cultivation annually. At a pricing of INR 59 per acre per month (INR 708 per acre per year), the Total Addressable Market (TAM) for Stoma Sense is estimated at INR 259.84 crore per year. Considering near- to medium-term deployment across organized sugarcane belts, irrigation command areas, and farms linked to sugar factories and digital extension programs, approximately 50% of this acreage is considered serviceable, resulting in a Serviceable Available Market (SAM) of about INR 129.9 crore per year. Based on phased adoption through pilots, factory-led rollouts, and government-supported programs, a conservative 10% penetration of the SAM is achievable over 3–5 years, translating to a Serviceable Obtainable Market (SOM) of approximately INR 13.0 crore per year.

For all Crops: Stoma Sense is a crop-agnostic remote sensing and GEO-AI platform that can be extended beyond sugarcane to all major crops cultivated in Maharashtra. The state's net sown area across all crops is approximately 42.74 million acres. At a pricing of ₹59 per acre per month (₹708 per acre per year), this

translates to a Total Addressable Market (TAM) of ~₹3,026 crore per year. Considering practical applicability to priority crops such as sugarcane, cotton, soybean, pulses, maize, wheat, and rice—together accounting for ~60% of the net sown area—the Serviceable Available Market (SAM) is approximately ₹1,815 crore per year. A realistic government-led pilot covering 5–10% of this SAM across selected districts would represent a Serviceable Obtainable Market (SOM) of ~₹90–180 crore annually, while creating a scalable digital agriculture foundation for statewide rollout.

Current Customers/ Pilots (if any)

14,000 Ha monitored for Grupo Porres in Mexico: Remote sensing, weeds, harvest route, sugarcane freshness- CAC in Dominican Republic 50,000 Ha monitored for Pantaleon in Guatemala: Remote sensing, weeds, harvest route, sugarcane freshness Results from Grupo Porres: a. Porres Group: Sugarcane Detection: Analysis of the group's plantings showed high accuracy in identifying areas that were not planted, while reports indicated that planting had occurred (98% accuracy between the detection model and field samples). b. Porres Group: Sugarcane Freshness: Potential savings of \$50,000 for each hour of freshness reduced between burning and milling. c. Porres Group: Fusarium Detection: An alarm event triggered the spread of a critical sugarcane disease, allowing for the implementation of sanitary barriers and preventing its spread to other sugarcane seed fields. Infected seed fields were identified, and the decision was made to burn them to prevent further spread. d. Porres Group: Weed Detection: The AI weed detection model supported application scheduling and improved weed control and effectiveness, with estimated savings of \$35,000.

Strength of Business Model

Sustainability of the Business

Stoma Sense is designed as a scalable, platform-based solution with low marginal cost and high reusability across geographies, customers, and crop types. The core architecture relies on satellite imagery, weather feeds, and farm-specific data—ensuring no hardware dependency and eliminating the need for costly on-ground installations. This allows the platform to be deployed rapidly across districts and states with minimal operational overhead. The deep-learning models, cloud-removal engine, and GEOAI prediction framework are built to generalize across clients, meaning the same training pipelines can be applied to new geographies with only incremental calibration. Crop-specific intelligence is incorporated through modular algorithms, enabling the platform to scale beyond sugarcane into other crops by embedding appropriate phenology, growth curves, and management practices. Because the platform is cloud-native and multi-tenant, it supports a wide customer base—farmers, FPOs, cooperatives, mills, and government programs—while maintaining cost efficiency. As more farms are onboarded, the model's accuracy improves through data network effects, further strengthening long-term sustainability. This combination of platform reusability, low operating cost, cross-crop adaptability, and geographic scalability ensures that Stoma Sense can grow sustainably while delivering consistent value to diverse agricultural markets.

Go-to-market Strategy

Our go-to-market strategy leverages international experience, strategic partnerships, and a scalable GEOAI platform. Deployments in Central and North America with Grupo Porres and Pantaleon provide strong credibility as we expand into India through pilots and research collaborations. A key differentiator is that Stoma Sense requires no hardware and runs entirely on satellite and weather data. Its deep-learning cloud-removal engine enables uninterrupted monitoring during monsoon, supporting reliable growth curves, HYDRO+ moisture insights, and predictive harvest indicators that standard NDVI tools cannot deliver. The platform's modular, multi-tenant architecture allows rapid, cost-efficient scaling across regions. AI/ML pipelines can be adapted to new geographies and crops with minimal re-training, giving Stoma Sense a low marginal cost and much faster deployment cycle than hardware-dependent competitors. We scale through institutional channels—FPOs, cooperatives, mills, NGOs, and government programs—supported by local partners who understand agronomy, language, and culture. Field engagement and change-management ensure adoption and generate feedback that improves the models. This combination of global traction, hardware-free scalability, cloud-free satellite

intelligence, and crop-adaptive algorithms positions Stoma Sense as a differentiated solution ready for district- and state-level expansion.

Please explain your pricing strategy. Additionally, describe how affordable your solution will be for farmers and how you have assessed their ability to pay

Stoma Sense uses a subscription model of ₹59 per acre per month (₹708 per acre annually), keeping costs low for small farmers while enabling large-scale adoption through FPOs, cooperatives, mills, and government programs. Affordability was benchmarked against Maharashtra sugarcane economics, where yields typically range from 80–100 tonnes per hectare (82–40 tonnes per acre). At the 2025–26 FRP of ₹3,550 per tonne, farmers earn roughly ₹1.1–₹1.4 lakh per acre annually, with some reporting up to ₹3 lakh.

Against this, the annual cost of ₹708 per acre is just 0.57% of gross revenue—far below typical losses from suboptimal irrigation, delayed stress detection, or poor harvest timing. In a 14,000-hectare field test, the platform delivered 100× ROI through avoidable-loss savings even after service costs.

Affordability is especially important given current FRP delays in Maharashtra, where only ~53% of dues were paid by November 2025 and arrears exceeded ₹1,980 crore. With farmers facing cash-flow uncertainty, a low-cost digital advisory that stabilizes yields and reduces risk becomes critical. The subscription remains below the cost of a single irrigation cycle, ensuring adoption even in delayed-payment years.

Because Stoma Sense has no hardware dependency, low marginal cost, and scalable GEOAI pipelines, the pricing model remains sustainable for district- and state-level expansion while addressing a TAM of ₹259.84 crore for sugarcane and ₹3,026 crore across major crops.

How many farmers, farmer groups, or land plots do you plan to include in your pilot, and what are your scale targets for expansion (e.g., number of farmers, area covered in acres or hectares)?

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Workplan & Milestones (monthly/quarterly)

Months 0–2: Setup & Data Integration Confirm pilot geography, mills and plots. Ingest historical yield, moisture, irrigation and variety data; validate polygons. Configure Sentinel-1/2 and weather ingestion, and deploy the cloud pipeline with basic dashboards. Milestone: Core data onboarded; infra live.

Months 3–4: Cloud-Free & Spectral Foundations Train the U-Net GAN cloud-free model and generate stabilized NDVI/NDRE/NDWI series. Implement pixel-to-plot aggregation and begin HYDRO+ and PGI feature extraction. Milestone: Clean spectral stack + PGI baseline.

Months 5–6: Moisture, HYDRO+ & PGI Build the cane-moisture model, calibrate HYDRO+ thresholds and refine PGI curves. Conduct limited field sampling to verify moisture %, weeds and growth stage. Milestone: Moisture + PGI models validated.

Months 7–8: Monitoring & Advisory Rollout Launch NDVI+, HYDRO+, PGI, weed maps and stress alerts. Release the mobile app with local-language advisories. Validate predictions via field visits; train agronomy and FPO teams. Milestone: Live advisory system active.

Months 9–10: Yield & Operational Intelligence Deploy the TCH yield model, publish plot-level maps and activate anomaly detection for stagnation, moisture shocks and weeds. Integrate insights into operational dashboards. Milestone: Yield model operational.

Months 11–12: Evaluation & Scale Plan Assess accuracy and agronomic impact (moisture-deficit reduction, TCH improvement, timely interventions). Milestone: Pilot validated; ready for expansion.

Risk Assessment & Mitigation (technical, operational etc)

1. Technical Risks Risk: Persistent cloud cover may reduce optical data quality. Mitigation: Use U-Net GAN cloud-free reconstruction, fuse Sentinel-1 radar (VV/VH), and maintain cached composites. Risk: Inconsistent

availability from weather/satellite APIs. Mitigation: Add retry logic, async queues, time-based fallbacks, and maintain local replicas of processed indices. Risk: Model drift due to seasonal, varietal or irrigation changes. Mitigation: Monthly retraining, monitoring of distribution shifts and automated recalibration alerts.

2. Data Quality & Governance Risks Risk: Incorrect or outdated plot boundaries, varieties or yield records. Mitigation: Automated geometry cleaning, field verification via mobile GPS and model weighting to reduce noisy-label impact. Risk: Data privacy issues under DPDP 2023. Mitigation: Encrypt data in transit and at rest, use anonymised IDs, and enforce strict RBAC with activity logs.

3. Operational Risks Risk: Delayed field sampling (moisture %, crop cuts, weeds). Mitigation: Train multiple officers, standardize capture through mobile workflows; allow photo-based remote validation. Risk: Low app adoption or limited field engagement. Mitigation: Simple UI, vernacular content, regular training & f/b, plus demonstration of early wins.

4. Infrastructure Risks Risk: High compute load during monsoon ingestion cycles. Mitigation: Auto-scaling microservices; scheduled parallel batch processing. Risk: Rural connectivity gaps. Mitigation: Offline caching for 48-72 hours

Training & Capacity Building Plan (Govt staff/SAUs/FPOs/Farmers)

1. Government & SAU Orientation (Months 1–2) Workshops for agriculture department, ATMA, and SAU staff on using the Stoma Sense platform, interpreting indices (NDVI+, HYDRO+, growth-curve deviations), and integrating insights into advisories. SOPs for data flow and monitoring established.
2. FPO Capacity Building (Months 2–4) Training FPO managers and extension leaders to onboard farmers, verify field boundaries, interpret predictive insights, and support digital adoption. Creation of “Digital Agri Champions” within participating FPOs.
3. Field Team Training (Months 2–6) Hands-on training for field monitoring officers and agronomy assistants on sampling standards, mobile data collection, geotagging, and weekly coordination with the Stoma Sense technical team.
4. Farmer Training & Demonstrations (Months 3–9) Village demonstrations and seasonal training sessions to help farmers interpret growth curves, moisture alerts, stress signals, and harvest forecasts. Continuous WhatsApp-based micro-learning and structured feedback loops.
5. Continuous Support & Change Management (Months 3–12) Rural digital facilitators provide on-ground assistance, conduct regular check-ins with farmers and FPO staff, and help refine advisory formats based on field feedback.
6. Knowledge Sharing & Scale Readiness (Months 10–12) Joint review sessions with FPOs, SAUs, and government stakeholders; documentation of learnings; and preparation of training manuals for expansion to additional districts.

Gender and Social Inclusion Plan

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Number of farmers expected to be impacted by the end of the pilot.

50

Output expected through this project.

1. Remote-Sensing Crop Intelligence Layer Generation of satellite-derived vegetation indices (NDVI, NDRE) for all mapped fields within the 100 sq km pilot zone. Development of enhanced, context-aware indices (NDVI+, HYDRO+, Smart Growth Index) through integration of remote sensing data with crop-stage information and farm-level attributes. Evaluation of higher-resolution, more granular satellite data to determine suitability for Indian field sizes and mixed cropping environments.
2. Predictive & Forecasting Models Field-level plant moisture predictions using satellite and harvest data. Forecasts for TSH, crop maturation trajectory, and harvest timing, calibrated to local climatic and agronomic conditions. Integration of any available historical harvest datasets from ADEX (or other local sources) to begin building and training ML models tailored to the region.
3. Data Quality Assessment & Model Readiness A systematic evaluation of data availability, completeness, and quality in the pilot region, including satellite coverage consistency, cloud-free imagery performance, and reliability of historical crop data. Identification of data gaps and requirements needed for full-scale predictive modeling and accurate advisory generation. Development of a regional data-readiness profile to inform future scaling across Maharashtra.
4. Farmer-Facing Decision Support Participating farmers (n=150) will receive individualized insights on crop growth, moisture conditions, and harvest-related forecasts. Collection of farmer feedback on how these

insights translate into real decisions (e.g., irrigation adjustments, scheduling of operations, early stress management). 5. Validation & Learning for Scale-Up Field validation of indices, predictions, and advisory workflows with support from the partnering KVK. Documentation of technical performance benchmarks, operational learnings, and requirements for district-level expansion.

Please specify the baseline values relevant to your project's focus area, the target values you aim to achieve, and the methods you will use to measure these values. Ensure the baseline you provide is directly related to the area or value your solution intends to impact.

There is no operational, high-frequency, cloud-free satellite monitoring system in the pilot region. While studies in Maharashtra and other sugarcane areas show NDVI/NDRE can track crop condition, these remain isolated research efforts, not tools available to farmers or extension staff. Farmers also lack enhanced indices such as NDVI+, HYDRO+, or phenology-based growth deviations that integrate crop-stage and field-specific information. Existing mill records provide harvest dates and cane weight, but no plot-linked historical dataset connects satellite time series with agronomic variables for TSH forecasting or harvest prediction. There are no digital decision-support tools that allow farmers to visualize crop status or anomalies, and the data quality and availability for the region (satellite coverage, historical records, moisture data) have not been assessed. This leaves the pilot area without a structured, analytics-driven foundation for irrigation, stress detection or harvest planning. Target (End of Pilot) • Full remote-sensing coverage of the 100 sq km pilot area, with validated NDVI/NDRE and enhanced indices. • Functional moisture prediction and early maturation/harvest forecasts for participating farms. • Regional dataset integrating satellite, weather and available ADEX harvest records assembled and evaluated. • Improved farmer situational awareness, with 50 farmers able to view crop-stage insights for their own fields. • Established data-readiness profile and validated technical workflow to support district-level scale-up. Measurement Method Remote-sensing outputs—indices, moisture predictions and forecasts—validated against field observations, farmer feedback and KVK agronomist assessments. Data quality assessed using satellite-coverage analytics, completeness of ADEX historical data and cloud-free model accuracy. Farmer-level outcomes measured through surveys.

Farmer centric Benefits (For e.g Increase in income, hour saved , accuracy, Improved Productivity, Yield Improvement, etc).

Stoma Sense provides farmers with timely, plot-level advisories on irrigation, crop stress, weed risk, and harvest readiness, improving decision accuracy and reducing guesswork. This leads to better yield stability, improved productivity, and higher and more predictable farm income. The solution enables efficient water and input use, lowering irrigation, energy, and labour costs, while early stress and weed detection helps prevent yield loss. Mobile-based, local-language advisories reduce the need for manual field inspections, saving time and effort for farmers. Overall, the pilot supports yield improvement, cost reduction, and sustainable income growth.

Environmental & Social Outcomes (For e.g., reduced carbon footprint, employment generation, etc).

Stoma Sense contributes to reduced water consumption and lower carbon footprint by enabling precise irrigation scheduling, minimizing over-irrigation, and reducing electricity and diesel use for pumping. Optimized input application and early stress detection help lower unnecessary agrochemical usage, supporting soil health and environmental sustainability. Socially, the pilot generates local rural employment through field monitoring, agronomy support, digital facilitation, and irrigation maintenance roles. Local-language mobile advisories improve digital inclusion, strengthen agri-extension systems, and empower farmers with data-driven decision-making, contributing to climate-resilient and sustainable agricultural livelihoods.

Core Team and Leadership

Name	Role	Qualification	Experience (- years)	LinkedIn URL	Expertise (Eg AI/ML, Agronomist, etc)
Dr Naina Bandyopadhyay	This role leads the end-to-end AI/ML pipeline, transforming satellite-, weather, and field data into reliable agronomic intelligence for farmers and institutions. They will also drive continuous model refinement, manage MLOps workflows, and ensure that analytical outputs directly support operational decision-making.	PhD	20	https://www.linkedin.com/in/10008377/	AI/ML
Sanket Pathak	The person will architect the cloud environment that powers Stoma Sense, optimizing compute, storage-, and dataflow pipelines for reliability and cost efficiency. They will also lead DevOps operations-, enabling continuous deployment-, system observability, and automated recovery to keep all services running smoothly in production.	MS in Computer Science	10	https://linkedin.com/in/sanket-pathak-dv	Cloud Architect-, DevOps
Chaitanya Rai	The person will own the end-to-end AI/ML and business analytics stack, including MLOps pipelines that keep models accurate, automated, and produc-	BTech	4	https://www.linkedin.com/in/chaitanya-rai-369881102?utm_source=-share&utm_cam-	MLOps

tion-ready. They will also translate analytical outputs into actionable advisories and measure impact to guide scale-up and government adoption.		paign=share_-via&utm_content=profile&utm_medium=ios_app			
Pranav Dhakulkar	The person will own the AI/ML and business analytics components of the project, building models and dashboards that convert raw data into practical advisory intelligence. They will also evaluate outcomes and ensure the analytical rigor needed for scale-up and government adoption.	PGP, MBA	7	https://www.linkedin.com/in/pranav-dhakulkar-6628a4342?utm_source=share&utm_campaign=share_via&utm_content=profile&utm_medium=ios_app	AI/ML, Business Analytics
Gaurav Sharma	The role focuses on implementing and optimizing machine learning components, including spectral feature engineering, anomaly detection, and calibration of field-level indicators. They will work closely with the AI/ML Lead to test model variants, document findings, and integrate improvements into the operational pipeline.	M.Sc.	2	https://www.linkedin.com/in/gaurav-sharma-957a14263/	AI/ML
Ratul Banik	The role focuses on exploratory modeling and error analysis, helping identify where models underperform and proposing targeted improvements. They	M.Sc.	1		AI/ML

will run ablation studies, analyze field validation results, and support the integration of refined algorithms into the production workflow.		https://www.linkedin.com/in/ratul-banik/	
Radhika Krishnaswamy	This role will provide senior oversight across the technical architecture and business strategy of the pilot, ensuring that GEOAI capabilities align with operational needs, stakeholder priorities, and long-term scalability. They will guide cross-functional teams, manage stakeholder expectations, and translate complex technical outputs into clear value pathways for mills, government partners, and farmers.	MBA	20 https://www.linkedin.com/in/radhika-krishnaswamy-667711/ Business and Computer Science

Advisor/Mentor for this project

Name	Qualifications	Expertise (Eg AI/ML, Entomologist, etc)	Experience (- years)	LinkedIn URL	Strategic Partners (if any SAU or Institution is a part of advisory)
					MGM's College of Agricultural Biotechnology, Aurangabad
					Baramati Agricultural University
					University of Kentucky, USA
					University of Tennessee, USA

File Upload

https://api-legacy.accubate.app/document/aiaic/552024c6-b792-49bc-80b8-bccd4e978ae0

Strategic impact on Maharashtra’s agriculture (policy alignment + scale potential + sustainability)

The proposed Stoma Sense pilot aligns closely with the Government of Maharashtra’s MahaAgri-AI Policy 2025–2029, which emphasizes farmer-centric adoption of AI, geospatial intelligence, and data-driven decision-making to improve productivity, climate resilience, and farmer income. The solution addresses a critical operational gap in agriculture by enabling continuous, plot-level crop monitoring even during the monsoon, when conventional remote-sensing tools are ineffective. Stoma Sense supports the state’s scale ambitions through a cloud-native, modular architecture that can be deployed across sugarcane belts and extended to other irrigated crops and agro-climatic zones with minimal configuration. This makes it suitable for phased, district-level and state-wide rollout through institutional and extension-led programs.

From a sustainability perspective, the platform enables efficient water use, early stress detection, and optimized input application, directly supporting Maharashtra’s priorities around water management, climate resilience, and sustainable agricultural growth. The pilot provides a replicable blueprint for scalable, AI-driven agricultural transformation across the state.

Funding Support required from AIAIC (in ₹ lakhs)

168

Activity wise funds utilization plan

Activity	AIAIC Support (₹ Lakhs)	Co funding(₹ Lakhs)	Total Cost (₹ Lakhs)
Hosting, Cloud Computing with third party Hyper-scalers like Microsoft, IBM or AWS	18	0	18
Human Resources. A team of 10 including Project Manager, Data Scientists, AI engineers, Ag experts, Data Engineers and training staff. Average cost of ₹2,50,000 per resource per month. This also includes resources based in United States and South America	150	150	300

Other Support required from the Government

Government farms for trialsGovernment Labs supportAccess to Agriculture Data Exchange Platform

We comply with applicable laws, DPIIT/startup norms (if applicable), and data privacy/security regulations.

Yes

We agree to periodic reviews, audits, and knowledge sharingwith Government/-SAUs.

Yes

We confirm the accuracy of information provided herein and ready to provide additional information if any required.

Yes

I/We agree to the above terms and conditions and authorize processing of this application as per the stated consent.

Yes